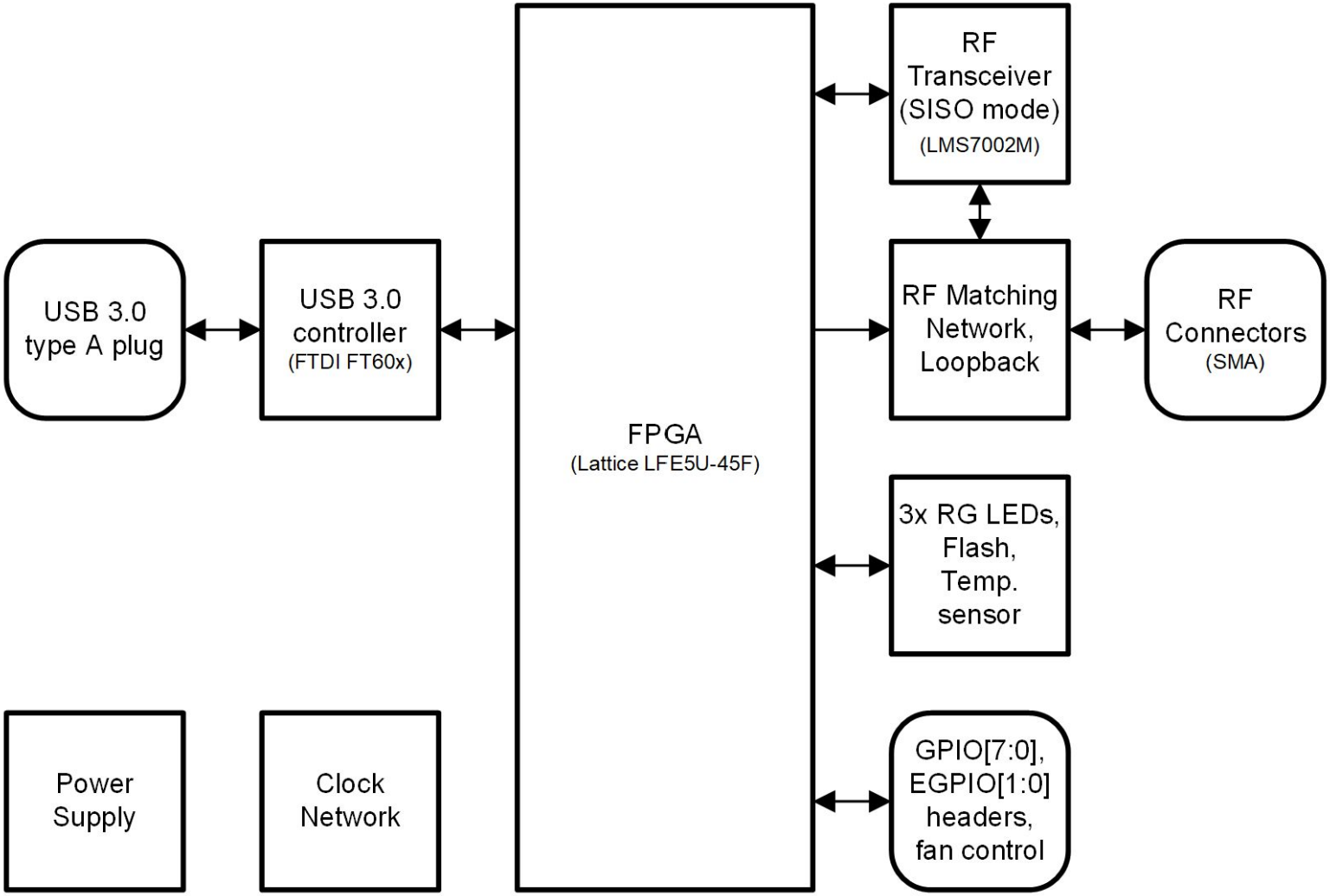
Block diagram

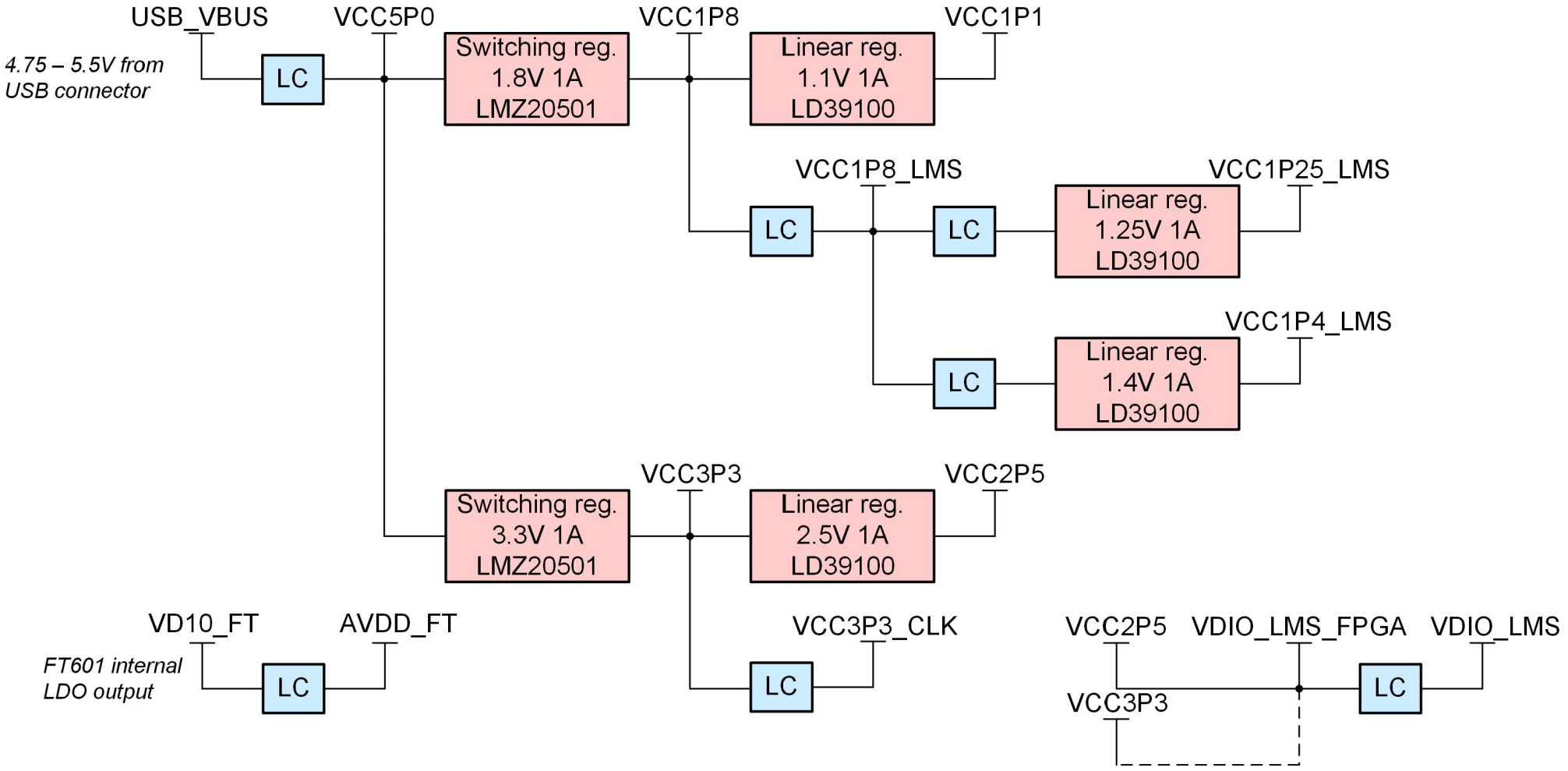


Project name: *LimeSDR-Mini_2v5.PrjPcb*			
Title: *Block diagram*			
Version: 2.5	Variant: Default		
Date: 10/14/2026	Time: 3:58:14 PM	Sheet 1 of 10	
File: 01_Block_Diagram.SchDoc		Size: A4	

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

Lime microsystems

Power diagram

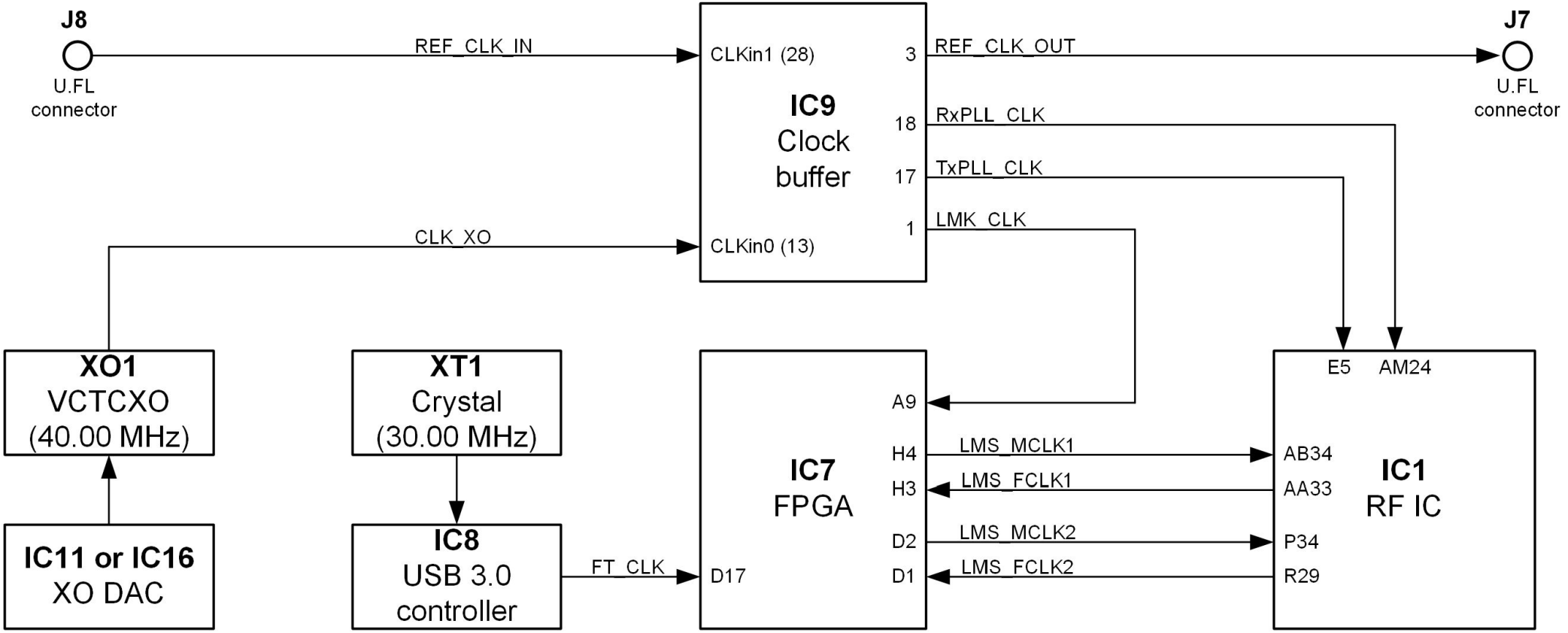


Project name:	*LimeSDR-Mini_2v5.PrjPcb*		
Title:	*Power diagram*		
Version: 2.5	Variant: Default		
Date: 10/14/2026	Time: 3:58:14 PM	Sheet 2 of 10	
File: 02_Power_Diagram.SchDoc		Size: A4	

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

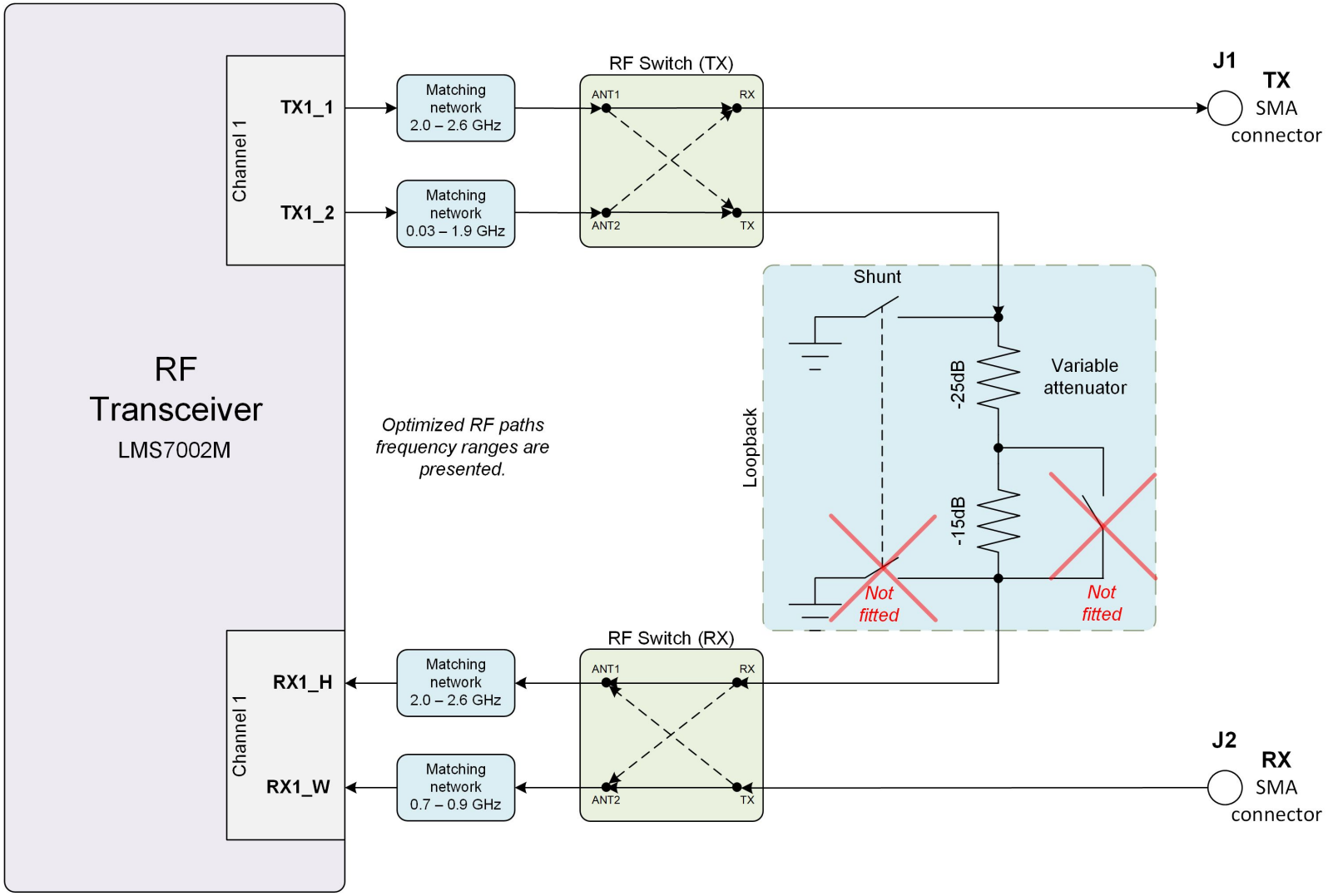
Lime microsystems

Clock diagram



Project name: *LimeSDR-Mini_2v5.PrjPcb*			
Title: *Clock diagram*			*Lime Microsystems* *Surrey Tech Centre* *Guildford GU2 7YG* *Surrey* *United Kingdom*
Version: 2.5	Variant: Default		
Date: 10/14/2026	Time: 3:58:14 PM	Sheet 3 of 10	
File: 03_Clock_Diagram.SchDoc		Size: A4	

RF diagram



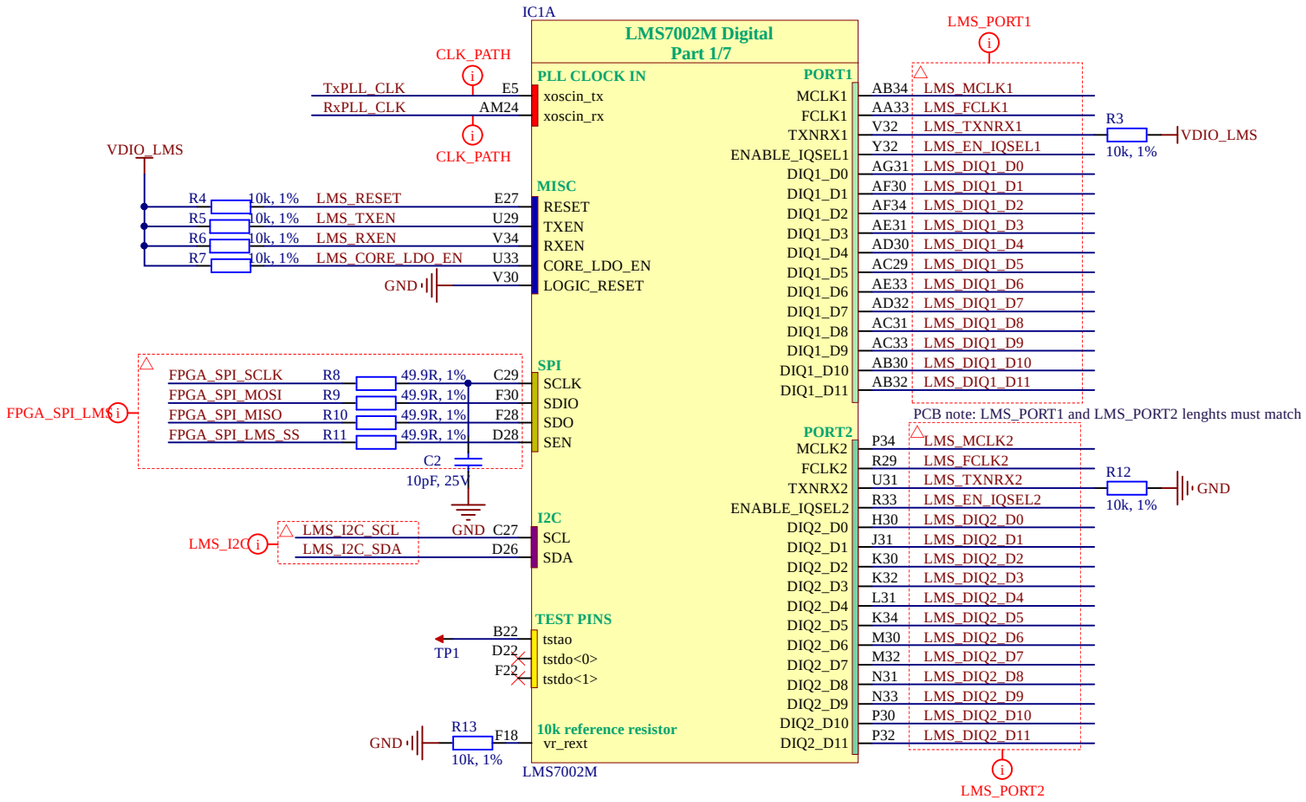
Project name:	*LimeSDR-Mini_2v5.PrjPcb*		
Title:	*RF diagram*		
Version: 2.5	Variant: Default		
Date: 10/14/2026	Time: 3:58:14 PM	Sheet 4 of 10	
File: 04_RF_Diagram.SchDoc		Size: A4	

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

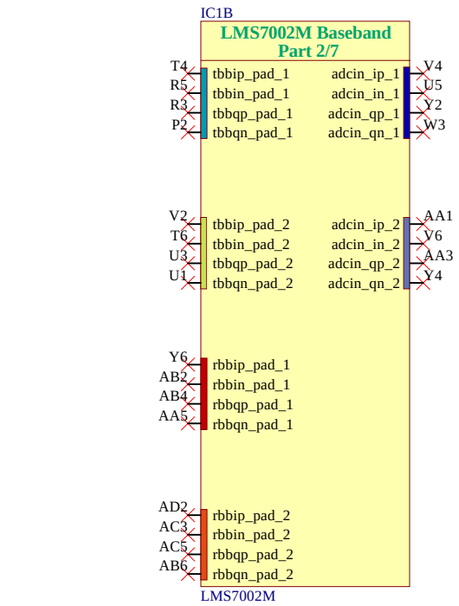
Lime microsystems

LMS7002M misc

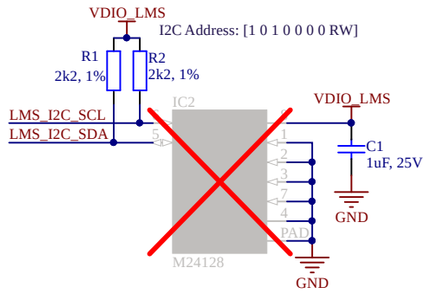
LMS7002M digital circuit



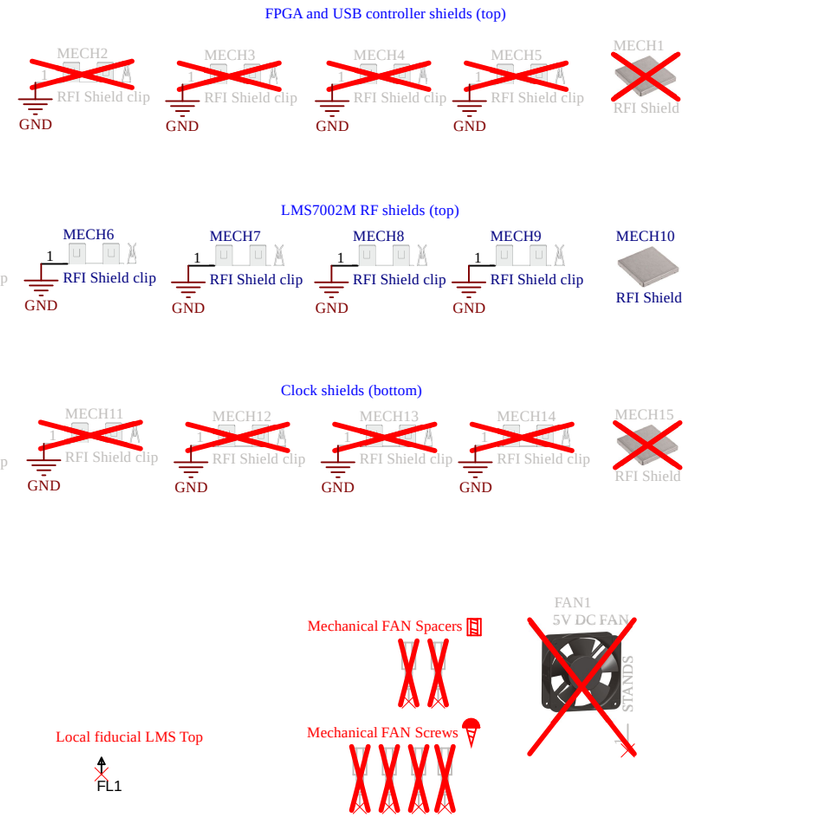
Baseband external IO



LMS EEPROM



Mechanical components

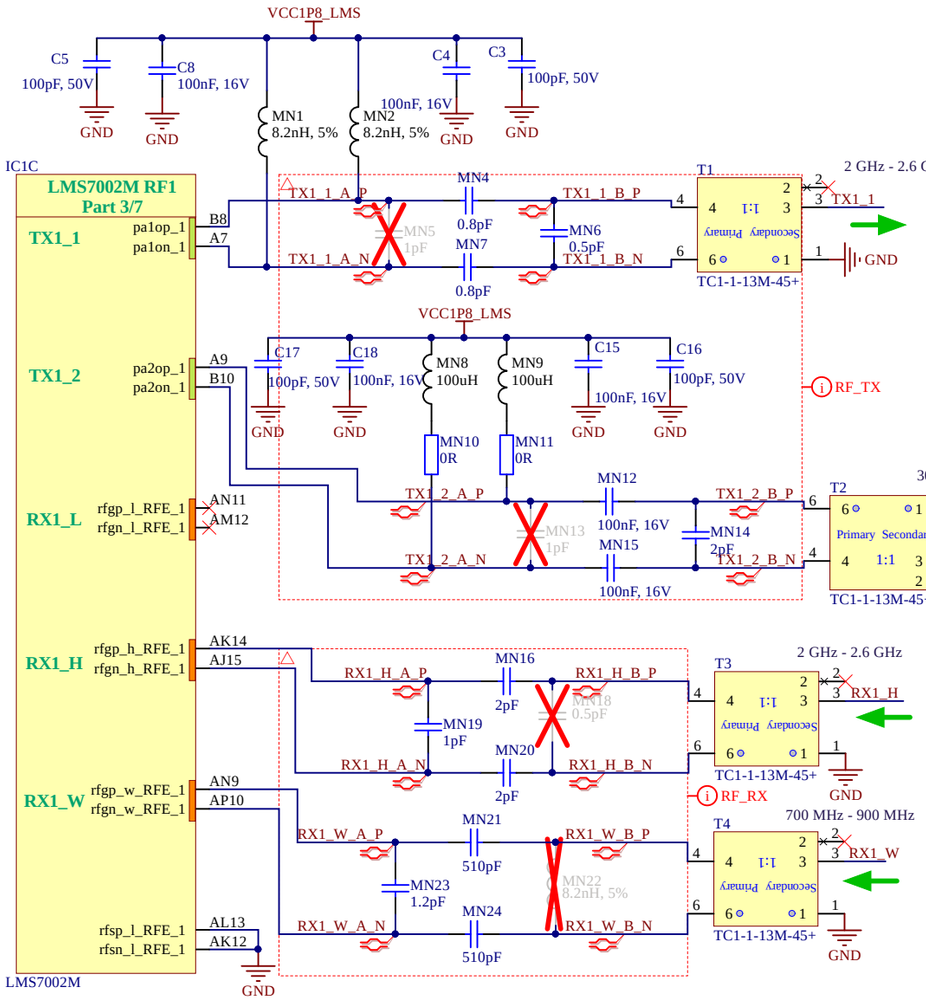


Project name: *LimeSDR-Mini_2v5.PrjPcb*			
Title: *LMS7002M misc*			
Version: 2.5	Variant: Default		
Date: 10/14/2026	Time: 3:58:14 PM	Sheet 5 of 10	
File: 05_LMS7002M_Misc.SchDoc		Size: A3	

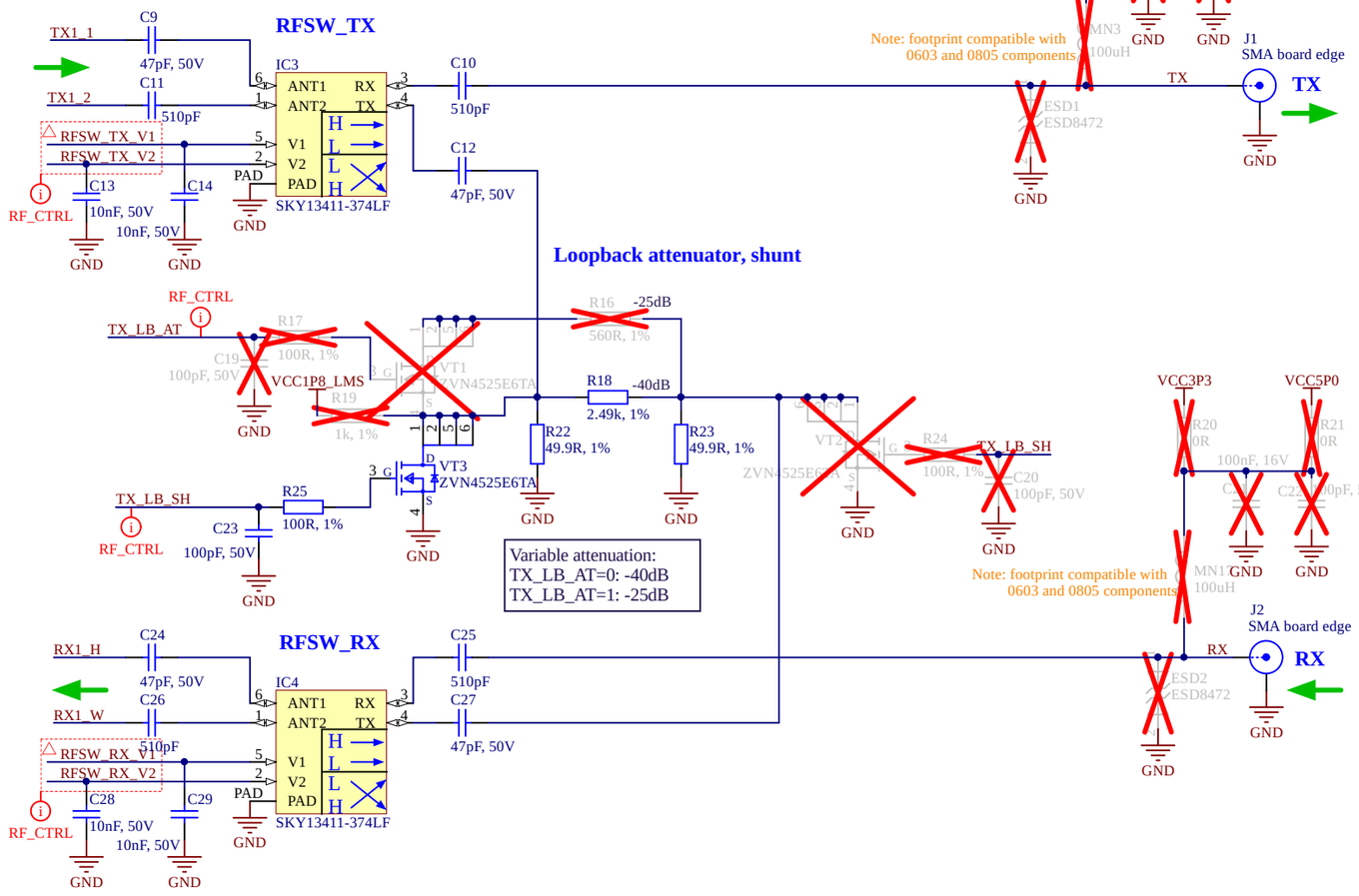
Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

LMS7002M RF circuits

LMS RF Channel 1

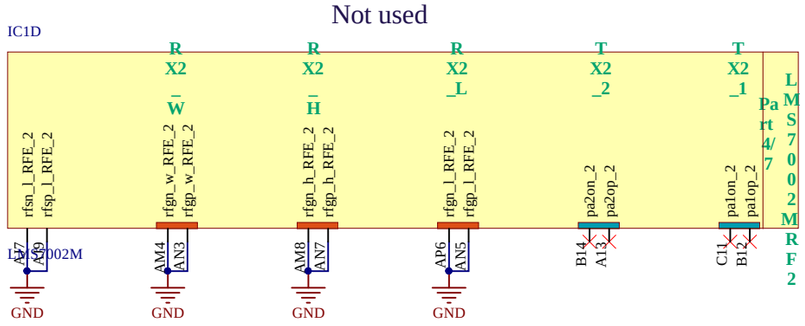


RF control



LMS RF Channel 2

Not used



RF truth table

RFSW_TX		RFSW_RX		J1	J2	Loopback
V1	V2	V1	V2	(TX)	(RX)	
H	L	H	L	TX1_1	RX1_H	TX1_2 → RX1_W
L	H	H	L	TX1_2	RX1_H	TX1_1 → RX1_W
H	L	L	H	TX1_1	RX1_W	TX1_2 → RX1_H
L	H	L	H	TX1_2	RX1_W	TX1_1 → RX1_H

RFSW (SKY13411) truth table

V1	V2	ANT1 (pin 6) → TX (pin 4)	ANT1 (pin 6) → RX (pin 3)	ANT2 (pin 1) → TX (pin 4)	ANT2 (pin 1) → RX (pin 3)
H	L	ISOLATION	ON	ON	ISOLATION
L	H	ON	ISOLATION	ISOLATION	ON

Project name:	LimeSDR-Mini_2v5.PrjPcb
Title:	LMS7002M RF
Version:	2.5
Variant:	Default
Date:	10/14/2026
Time:	3:58:15 PM
Sheet	6 of 10
File:	06_LMS7002M_RF.SchDoc
Size:	A3

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

LMS7002M power supply circuit

IC1F
LMS7002M Power Part 6/7
1.25V Digital
DIGPRVDD1 AE29
DIGPRVDD1 AA29
DIGPRVDD1 L33
DIGPRVDD1 R31
DVDD_SXR AL21
DVDD_SXT M4
VDD12_DIG C25
VDD_SPI_BUF F24
DVDD_CGEN A17
1.25V Analog
VDD12_TXBUF G5
VDD12O_VCO_SXT J5
VDD12_VCO_SXT K6
VDD_CP_SXT J3
VDD_TBB P4
VDD12_TIA_RFE AD6
VDD12_LNA_RFE AG5
VDD_CP_SXR AL19
VDD_DIV_SXR AM20
VDD12_VCO_SXR AN23
VDD12_RXBUF AP26
VDD_AFE AJ25
VDD_CP_CGEN F20
VDD_DIV_CGEN E19
VDD_TPAD_TRF B16
VDD_TLOBUF_TRF F8
VDDO_TLOBUF_TRF D6
1.25V-1.4V Analog
VDD_DIV_SXT L5
VDDO_DIV_SXT K2
VDD_MXLOBUF_RFE AN17
1.4V Analog
VDD14_RBB AD4
VDD14_TIA_RFE AF2
VDD14_LNA_RFE AF6
VDD14_VCO_CGEN C21
LMS7002M

Only one is fitted
VCC1P4_LMS
R26 0R
R27 0R
VCC1P25_LMS
600R, 0.27A FR1
600R, 0.27A FR2
600R, 0.27A FR3
C32 1uF, 6.3V
C33 1uF, 6.3V
C34 1uF, 6.3V
C35 1uF, 6.3V
C36 1uF, 6.3V
C37 1uF, 6.3V
C38 1nF, 10%
C39 1uF, 6.3V
C40 1nF, 10%
C41 1uF, 6.3V
C42 1nF, 10%
C30 1uF, 6.3V
C31 1nF, 10%
C43 1uF, 6.3V
GND
R28 10R
FR4 (fitted)
FR4 FR5 FR6 FR7 FR8 FR9 FR10 FR11 FR12 FR13 FR14 FR15 FR16 FR17
VCC1P4_LMS
All ferrites are 600R, 0.27A
R29 10R
C45 1uF, 6.3V
C46 1uF, 6.3V
C47 1nF, 10%
C48 1uF, 6.3V
C49 1uF, 6.3V
C50 1uF, 6.3V
C51 1uF, 6.3V
C52 1uF, 6.3V
C53 1uF, 6.3V
C54 1uF, 6.3V
C55 1uF, 6.3V
C56 1uF, 6.3V
C44 1uF, 6.3V
C57 1uF, 6.3V
C58 1uF, 6.3V
C59 1uF, 6.3V
GND
FR18 600R, 0.27A
FR19 600R, 0.27A
FR20 600R, 0.27A
FR21 600R, 0.27A
VCC1P4_LMS
C61 1nF, 10%
C62 1uF, 6.3V
C63 1uF, 6.3V
C64 1uF, 6.3V
C65 1uF, 6.3V
C66 1uF, 6.3V
C60 1uF, 6.3V
GND

IC1G
LMS7002M GND Part 7/7
Digital gnd: N3, AM22, E25, C23, B18, J33, AA31, AG29, T30, W29, G31, AF32, Y30
DGND_SXT, DGND_SXR, GND_SPI_BUF, GND_DIG, DGND_CGEN, DIGPRGND1, DIGPRGND1, DIGPRGND1, DIGPRGND1,2, DIGPRGND1, DIGPRGND2, DIGPRGND2, DIGPRGND2
Analog gnd: L3, K4, AN25, AK20, AJ19, B20, F6, D20, AL23, H2, C7
GND_DIV_SXT, GND_CP_SXT, GND_RXBUF, GND_DIV_SXR, GND_CP_SXR, GND_CP_CGEN, GND_TXBUF, GND_DIV_CGEN, GND_VCO_SXR, GND_VCO_SXT, GND_TLOBUF_TRF
Thermal pad EP: GND
LMS7002M
GND GND

IC1E
LMS7002M Power Part 5/7
1.8V Digital
VDD18_DIG D24
R30 0R
VCC1P8_LMS
C67 1uF, 6.3V
GND
1.8V-3.3V Digital
DIGPRVDD2 AH30
DIGPRVDD2 W33
DIGPRVDD2 T32
DIGPRVDD2 H32
DIGPRVDD2 W31
DIGPRPOC
VDIO_LMS
R32 0R
R34 0R
R35 0R
R36 0R
R37 0R
C72 1nF, 10%
C73 1uF, 6.3V
C74 1nF, 10%
C75 1uF, 6.3V
C76 1nF, 10%
C77 1uF, 6.3V
C78 1nF, 10%
C79 1uF, 6.3V
C80 1nF, 10%
C81 1uF, 6.3V
1.8V Analog
VDD18_VCO_SXT G3
VDD18_LDO_TX N1
VDD18_TIA_RFE AE3
VDD18_LDO_RX AF4
VDD18_SXR AM18
VDD18_VCO_SXR AK22
VDD18_BIAS E17
VDD18_TRF D16
VDD18_VCO_CGEN E21
VCC1P8_LMS
FR24 FR26 FR27 FR28 FR29 FR30 FR31 FR32
C86 1uF, 6.3V
C87 1uF, 6.3V
C88 1uF, 6.3V
C89 1uF, 6.3V
C90 1uF, 6.3V
C91 1uF, 6.3V
C92 1uF, 6.3V
C85 1uF, 6.3V
GND
1.8V-3.3V Analog
VDD18_TXBUF H6
VDD18_RXBUF AM26
FR33 FR34
R40 10R
R41 10R
VCC1P8_LMS
C93 1uF, 6.3V
C94 1uF, 6.3V
C95 1uF, 6.3V
C96 1uF, 6.3V
GND GND GND GND GND
LMS7002M

Linear regulators

VCC1P8_LMS
FR22 220R, 0.9A
C68 4.7uF, 6.3V
GND
IC5 LD39100PUR
Vin, Vout, EN, ADJ, GND, PAD, PAD, PG
VCC1P25_LMS
R31 8.2k, 1%
R33 15k, 1%
C69 4.7uF, 6.3V
C71 100pF, 50V
GND
VCC1P8
FR23 600R, 2A
VCC1P8_LMS
C70 10uF, 6.3V
GND

VCC1P8_LMS
FR25 220R, 0.9A
C82 4.7uF, 6.3V
GND
IC6 LD39100PUR
Vin, Vout, EN, ADJ, GND, PAD, PAD, PG
VCC1P4_LMS
R38 7.5k, 1%
R39 10k, 1%
C83 4.7uF, 6.3V
C84 100pF, 50V
GND

Project name:	LimeSDR-Mini_2v5.PrjPcb
Title:	LMS7002M power
Version:	2.5
Variant:	Default
Date:	10/14/2026
Time:	3:58:15 PM
Sheet	7 of 10
File:	07_LMS7002M_Power.SchDoc
Size:	A3

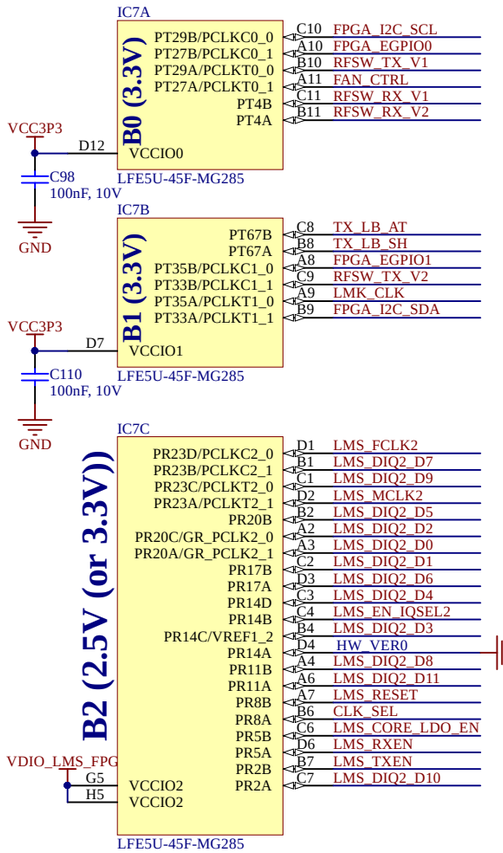
Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

Lime microsystems

FPGA

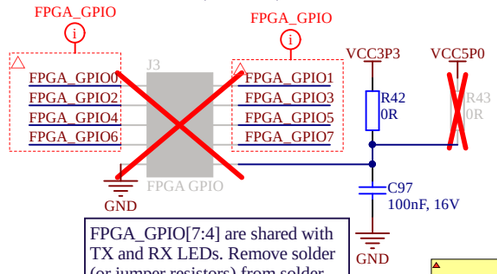
FPGA Banks

By default LFE5U-45F-xxMG285 is fitted and the schematic symbol represents LFE5U-25F pins. LFE5U-45F and LFE5U-85F parts are footprint and pin compatible with LFE5U-25F with small nuances. For example VCCIO0 bank C10 pin has different function:
LFE5U-25F: C10 pin is PT29B/PCLKC0_0
LFE5U-45F: C10 pin is PT38B/PCLKC0_0
LFE5U-85F: C10 pin is PT65B/PCLKC0_0
Therefore if a different to the default FPGA is used, one should look up the proper function note for each pin.
All power and fixed-function pins are the same for LFE5U-25F, LFE5U-45F, LFE5U-85F parts.



FPGA GPIO

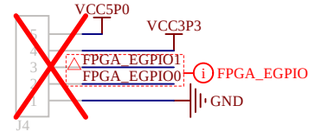
Pitch: 0.05" (1.27mm)



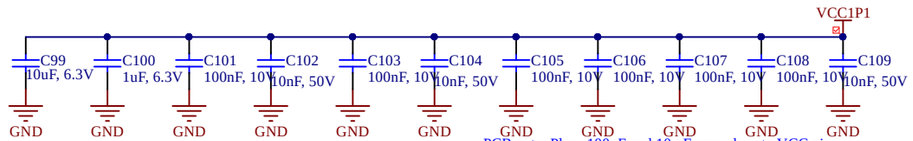
FPGA_GPIO[7:4] are shared with TX and RX LEDs. Remove solder (or jumper resistors) from solder bridges to disconnect LEDs from GPIOs lines.

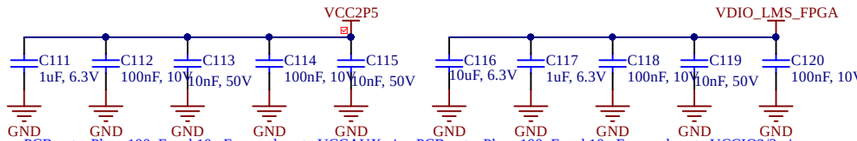
FPGA GPIO (edge)

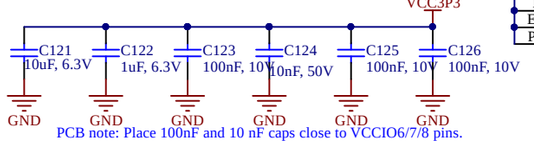
Pitch: 0.1" (2.54mm)

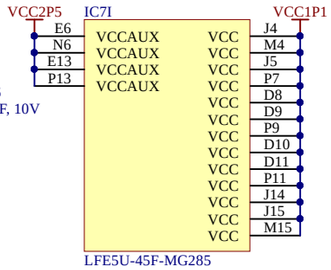


Decoupling

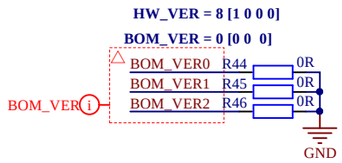




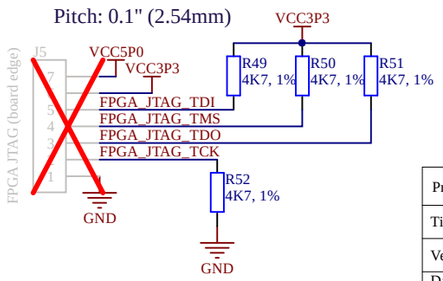


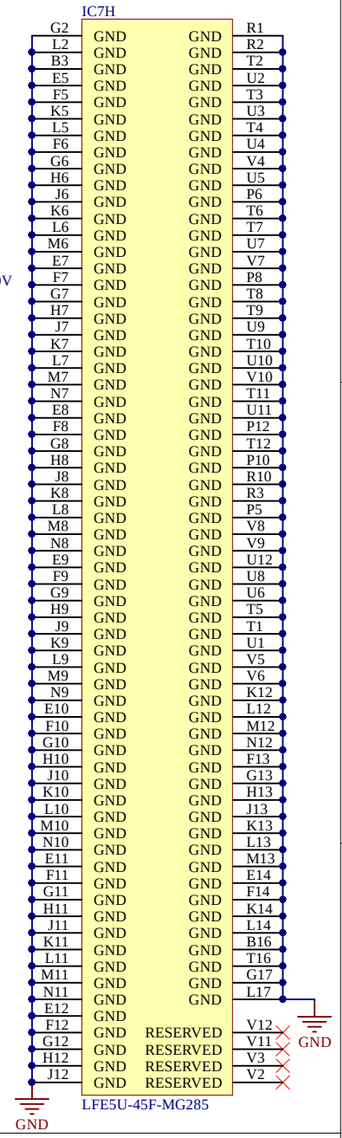


HW_VER & BOM_VER



FPGA JTAG (edge)





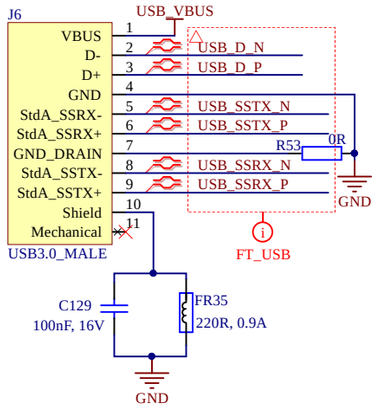
Project name:	*LimeSDR-Mini_2v5.PrjPcb*
Title:	*FPGA*
Version: 2.5	Variant: Default
Date: 10/14/2026	Time: 3:58:16 PM
File: 08_FPGA.SchDoc	Sheet 8 of 10
Size: A3	

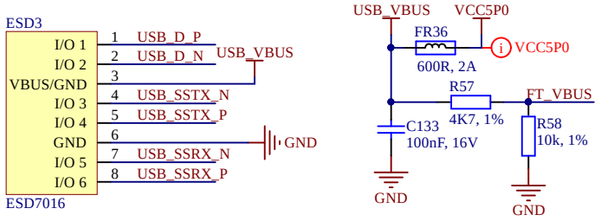
Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

Lime microsystems

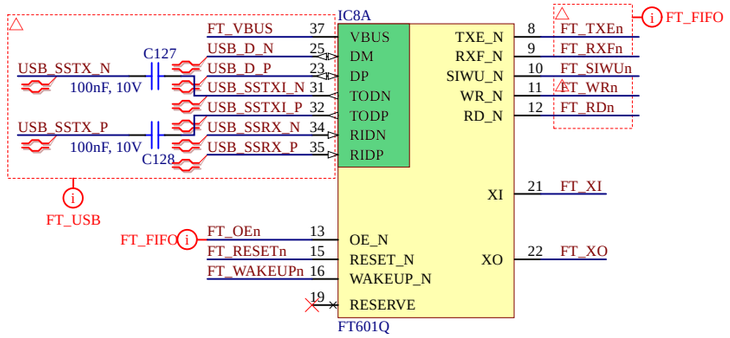
USB 3.0 to FIFO interface

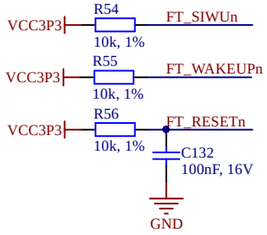
USB 3.0 type A plug and ESD

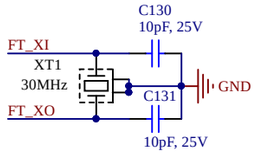


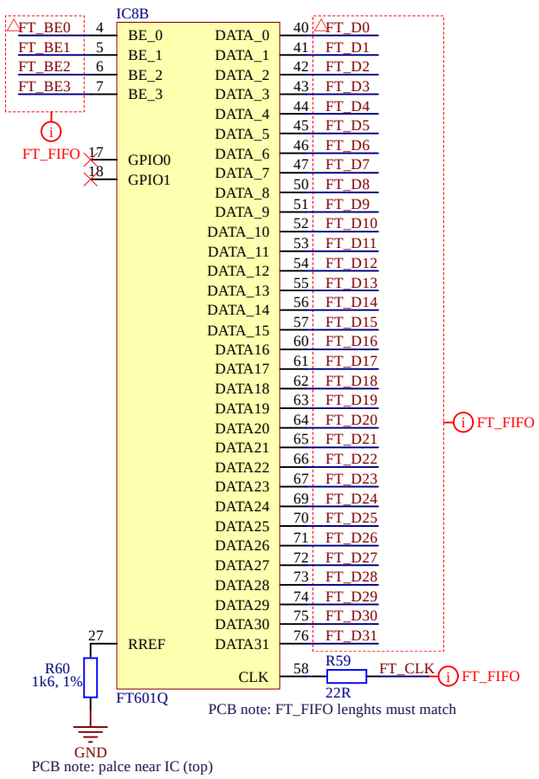


FTDI digital interface

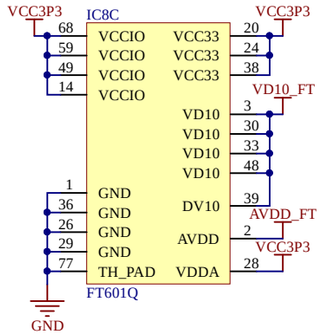


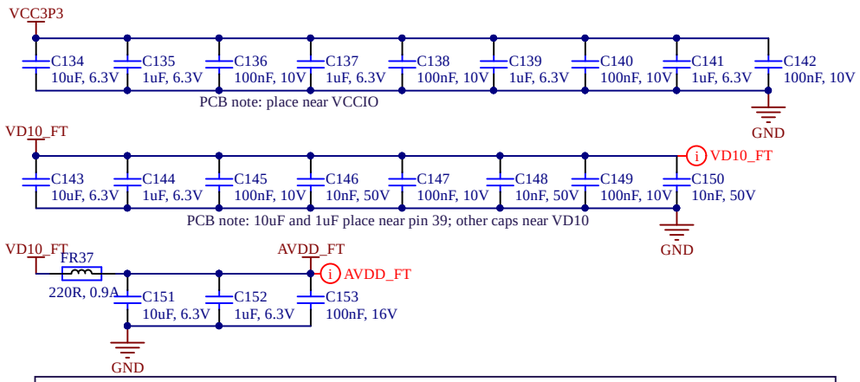






FTDI Power





Schematic note: marked 10uF caps (C134, C143, C151) not crucial and can be removed to reduce BOM price

Project name: *LimeSDR-Mini_2v5.PrjPcb*			
Title: *USB3.0 device*			
Version: 2.5	Variant: Default		
Date: 10/14/2026	Time: 3:58:16 PM	Sheet9 of 10	
File: 09_USB3_0_FIFO.SchDoc		Size: A3	

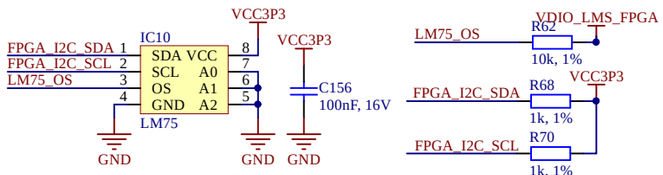
Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surray
United Kingdom

Lime microsystems

Misc

I2C Temperature sensor

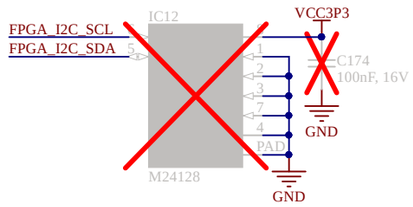
I2C addr: [1 0 0 1 0 0 0 RW]



PCB note: place between FPGA and LMS

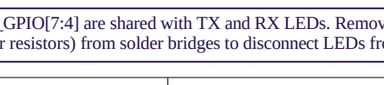
I2C EEPROM

I2C Address: [1 0 1 0 0 0 0 RW]

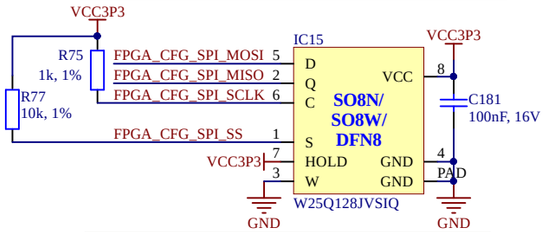


I2C secure key storage

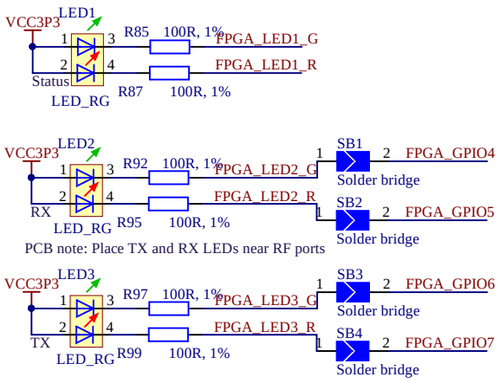
I2C addr: [1 1 0 0 0 0 0 R/W] (default)
I2C addr: [1 1 0 0 0 0 1 R/W] (alternative)
Avoid conflict with switching regulator



FPGA configuration Flash

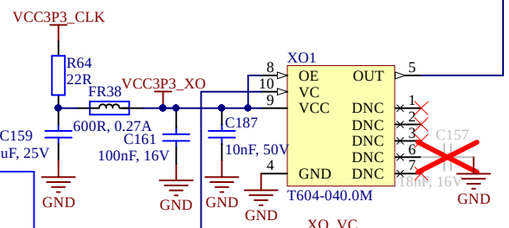


FPGA LEDs (Red/Green)



FPGA_GPIO[7:4] are shared with TX and RX LEDs. Remove solder (jumper resistors) from solder bridges to disconnect LEDs from GPIOs lines.

(VC)TCXO



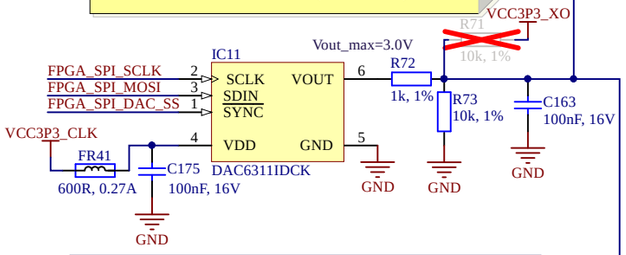
Frequency: 40 MHz
Stability: ±0.28 ppm

Output: LVCMOS
VOL: 0.1VCC max
VOH: 0.9VCC min

VCTRL: 0.3 .. 1.65 .. 3.0 V

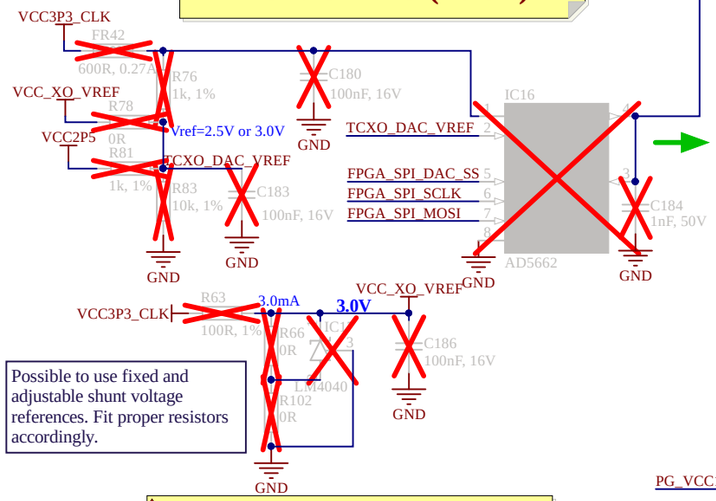
Optional capacitor for Rakon E7355LF pin 7 (Vfilter)

XO DAC (10 bit)



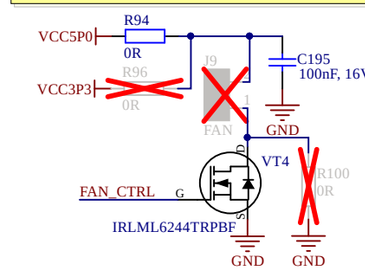
IC10 alternatives: DAC5311IDCK (8 bit) or DAC5311IDCK (12 bit). Compatible FPGA GW must be used.

XO VC DAC (16 bit)

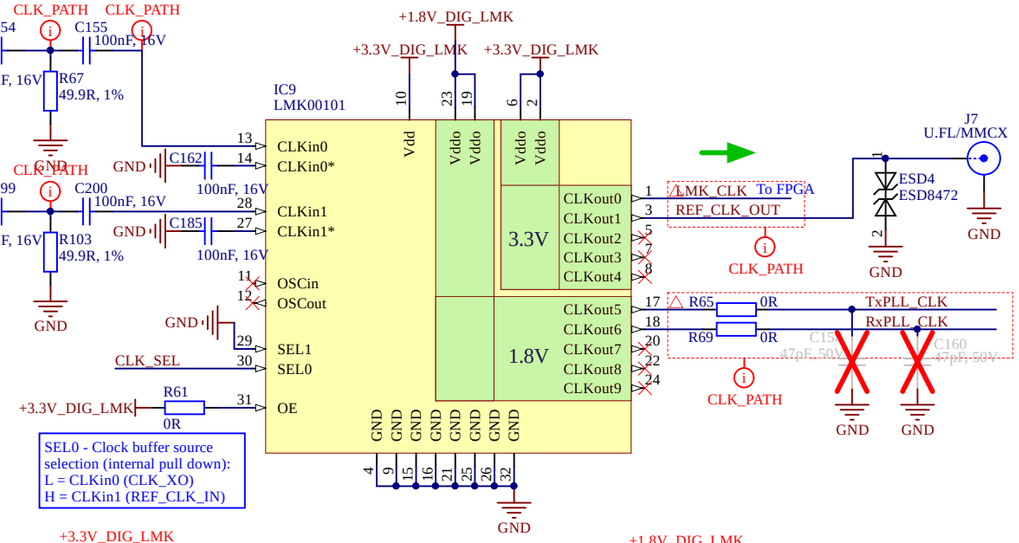


Possible to use fixed and adjustable shunt voltage references. Fit proper resistors accordingly.

FAN control

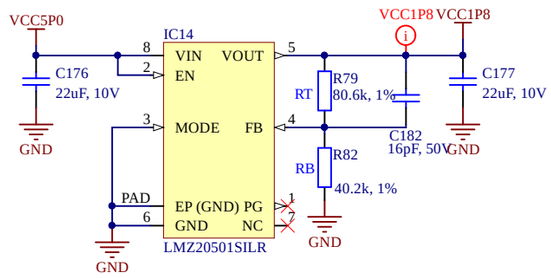


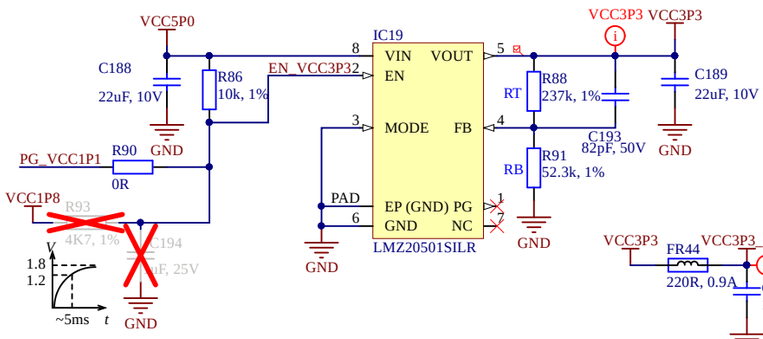
Clock buffer



SEL0 - Clock buffer source selection (internal pull down):
L = CLKin0 (CLK_XO)
H = CLKin1 (REF_CLK_IN)

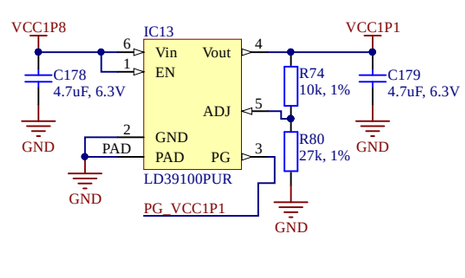
Switching regulators

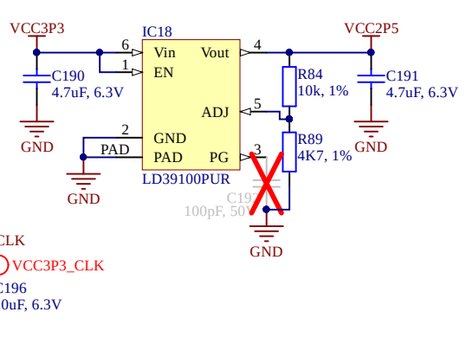




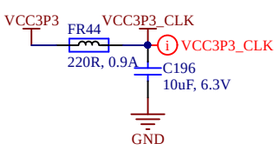
PCB note: Place 100pF close to each Vdd and Vddo pins.

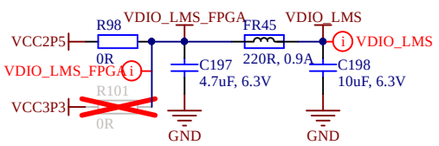
Linear regulators





PCB note: Place 100pF close to each Vddo pins.





Project name:	*LimeSDR-Mini_2v5.PrjPcb*
Title:	*Misc*
Version: 2.5	Variant: Default
Date: 10/14/2026	Time: 3:58:17 PM
File: 10_Misc.SchDoc	Sheet 10 of 10
	Size: A3

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

Lime microsystems